Choose a motion method

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Choose a motion relationship and justify its assumptions.

SUCCESS CRITERIA - I can...

1. Choose a relationship using the known quantities.
2. Show linked calculations with units and directions.
3. State when constant velocity or acceleration applies.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A cart starts at 3 m and moves right at 5 m/s for 2 s. Predict final position. Allowed endpoints are 11 to 15 m; state this interval and one assumption.

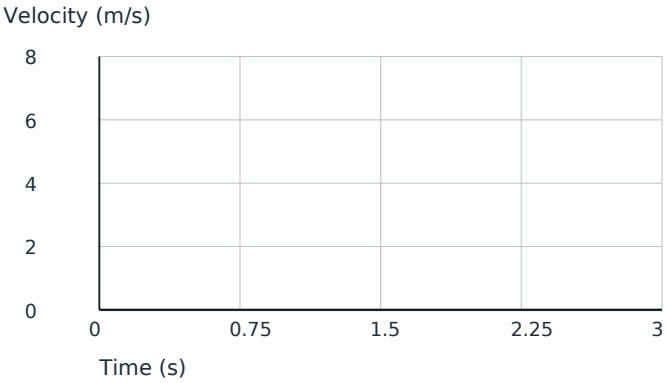
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $v_f = v_i + a\,\Delta t$; $\Delta x = \frac{1}{2}(v_i + v_f)\Delta t$. Constant acceleration; i = initial, f = final. At constant velocity, $\Delta x = v\,\Delta t$.

Worked reasoning: From rest, a cart accelerates uniformly at 2 m/s² for 3 s. Final velocity=6 m/s; average velocity=(0+6)/2=3 m/s; displacement=3×3=9 m.

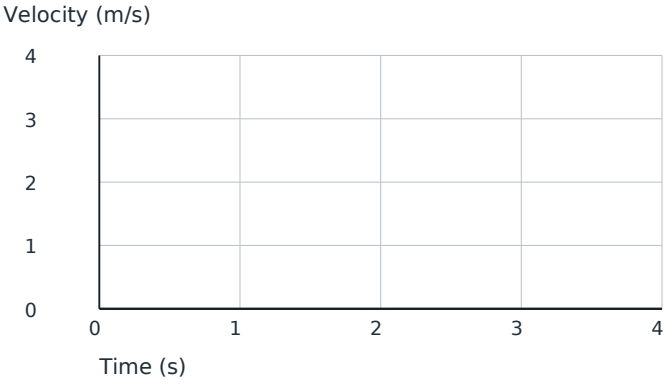
Your turn: Complete the cart's velocity-time graph. Shade the triangular area under the line and use it to check the 9 m displacement.



Marzano check - circle: ① ② ③

03 Choose the relationship

A robot drives straight at a steady 3 m/s for 4 s. Sketch its velocity-time graph. Decide whether a constant-velocity or acceleration relationship fits, then find its displacement.



Marzano check - circle: ① ② ③

04 Build final velocity

A remote-control car starts at 2 m/s east and gains 2 m/s each second for 3 s. Complete the table. How much velocity does it gain altogether, and what is its final velocity?

Time (s)	Velocity east (m/s)
0	2
1	4
2	6
3	?

Marzano check - circle: ① ② ③

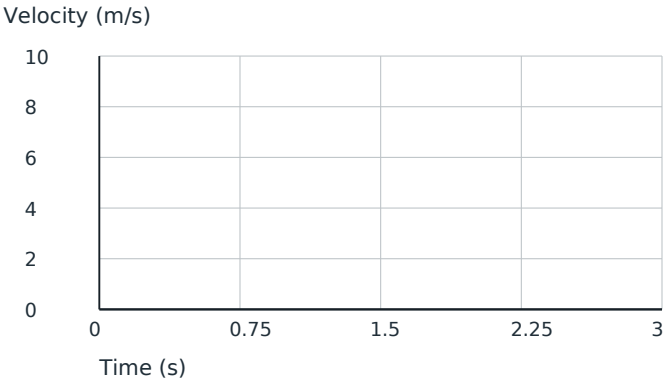
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Continue down each column.

05 Find the distance while speeding up

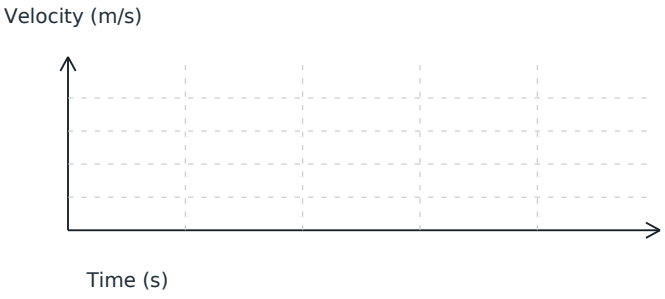
Use the car's velocities from question 4 to complete the graph. Its acceleration is constant. Find its average velocity and displacement over the 3 s. Use the graph's area to check.



Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

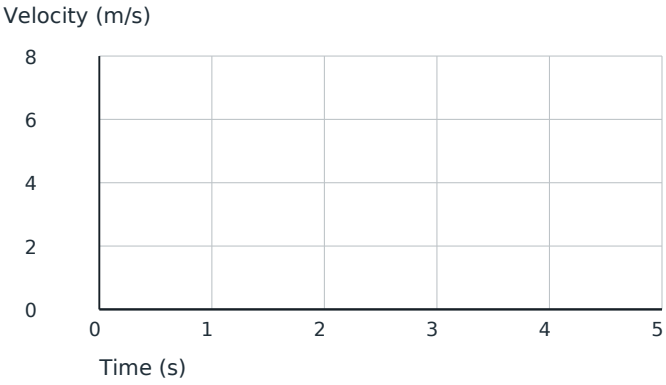
A cart travels west at 10 m/s and west at 3 m/s 2 s later. Is it speeding up? Find acceleration size and direction. Sketch the velocity-time model; label the given values.



Marzano check - circle: ① ② ③

07 Exit check

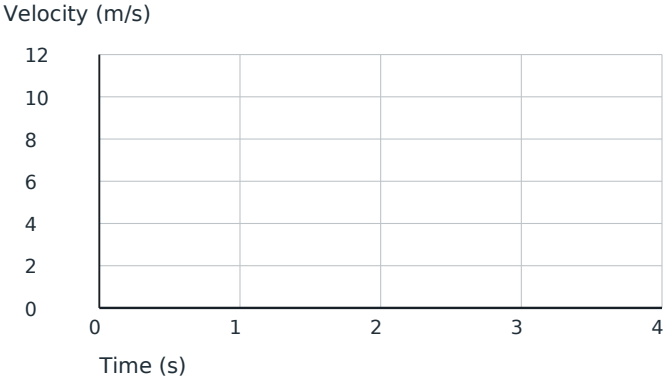
In a launch test, a small rover speeds up steadily from rest to 6 m/s east in 5 s. Plot the two endpoints and connect them. Find acceleration, average velocity and displacement.



Marzano check - circle: ① ② ③

08 STRETCH: test the model

Create two different motions that travel 8 m in 4 s: one at steady speed and one speeding up uniformly from rest. Draw their velocity-time graphs.



Marzano check - circle: ① ② ③